# Smart Signal: A Walk-Forward Validated Machine Learning Swing-Trading System for NSE Equities

Machine Learning (CSE_3148) Mini Project Report

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GitHub repository: [add link after pushing the code]

Date of submission: October 2026

# 1. Abstract

This project builds an end-to-end machine learning system that decides when to buy National Stock Exchange (NSE) equities for short swing trades, executes those decisions through the Fyers broker API in paper-trading mode, and presents everything on a custom web dashboard called ATLAS. A first version of the project framed the problem as three-class BUY/HOLD/SELL prediction on four stocks and reached only 37.3% test accuracy; in live use it bought two stocks and never sold them. The final version reframes the task as binary classification of trade outcomes: for every stock and every day, the label says whether buying at the next session's open and exiting with fixed take-profit, stop-loss and 15-day time rules would have made money after real brokerage charges. A bagged ensemble of three histogram gradient-boosted tree models is trained on 740,519 labelled stock-days covering 191 liquid NSE stocks from 2007 to October 2026, using 57 scale-free technical, market-regime and cross-sectional features. Evaluation uses a strict walk-forward protocol in which every test year is predicted by a model trained only on earlier data. On the 2022 to October 2026 holdout the system made 87 trades with an 87.4% win rate after charges, a profit factor of 2.04 and an average net return of +0.76% per trade; across all out-of-sample years from 2018 it made 133 trades with an 89.5% win rate. The report also discusses the main limitations honestly: the model's overall ranking ability is weak (ROC AUC about 0.53), the edge is concentrated in the rare highest-confidence setups, trading is infrequent, and results are sensitive to position size because of fixed brokerage fees.

# 2. Introduction

Retail algorithmic trading in India has grown quickly because brokers such as Fyers expose order, quote and historical data APIs to individual users. A common first approach is to combine technical indicators (moving averages, RSI, MACD and so on) into a hand-written score and trade when the score crosses a level. Such rules are easy to build but their weights are guesses, and they are rarely tested on data they were not designed on.

Machine learning offers a principled alternative: show a model many historical examples of indicator values together with what happened next, and let it learn the mapping. Financial time series, however, are noisy, non-stationary and heavily competed, so naive models easily look good in a random train/test split and fail in real use because of look-ahead bias, overlapping labels or ignored trading costs.

This project started from an existing rule-based Fyers bot and a first ML model (version 1). Version 1 predicted BUY/HOLD/SELL over five days for four stocks (HDFCBANK, AUBANK, BAJAJFINSV, RELIANCE) using five years of data. It reached 37.3% accuracy (macro F1 0.359), barely above the 33% chance level for three classes. In live paper trading it bought HDFCBANK and AUBANK but never sold, because the model never output SELL for a held stock and the bot had no other exit. Its simulation mode used randomly generated prices, so its reported win rate said nothing about the model. Version 2, described in this

report, redesigns the learning problem, the data, the validation procedure, the trading loop and the user interface.

## 3. Objective

1 Reformulate the trading decision as a supervised learning problem whose label matches what the bot actually does (enter, then exit by explicit rules), including real Fyers brokerage and statutory charges.
2 Build a large, clean, split-adjusted historical dataset of NSE stocks and market indices from public sources.
3 Engineer scale-free features, including the indicators shown on the ATLAS dashboard (SMA 20/50/200, EMA 20), so one model can learn across many stocks.
4 Train and select a model using strict walk-forward validation with an embargo, and choose the buy threshold without looking at the final test years.
5 Target an out-of-sample win rate of at least 85% after charges while keeping a positive profit factor.
6 Integrate the model into a live Fyers paper-trading bot with take-profit, stop-loss and time exits, persistent positions and robust handling of expired login tokens and API rate limits.
7 Integrate the ATLAS front end with a Python backend serving real market data, the model's view on every NSE stock, and three bot views: LIVE (the real market), SIM (the model's real out-of-sample trades replayed in real time on real NSE prices) and LIVE SIM (the bot trading a simulated market in real time, 24 hours a day, with no connection to past data).

## 4. Data Source

| Source | Content | Use |
|---|---|---|
| eod2 dataset (GitHub, BennyThadikaran/eod2_data) | NSE bhavcopy end-of-day OHLCV and delivery volume since 1995, adjusted for splits and bonuses; 191 stocks downloaded, plus NIFTY 50, India VIX and 10 sector indices | Training and backtesting history up to 13 October 2026 |
| Fyers History API | Daily candles for the same symbols and indices | Verification (matched eod2 to the paisa on 1,233 overlapping days per stock) and daily top-up |
| Fyers Quotes API | Live last-traded price | Live exits and entries in the bot |
| Fyers charges list (fyers.in) | Brokerage, STT, exchange, stamp duty, GST, DP fees | Cost model inside the label |
| Yahoo Finance (yfinance), Moneycontrol and Economic Times RSS, NSE | Global quotes, fundamentals, news, FII/DII flows | ATLAS website only (not used for training) |

**Preprocessing.** Data before 2007 was dropped because the source contains unadjusted corporate actions in earlier years. Each stock's history was split into separate segments at any

one-day move larger than 35% (demergers such as Adani Enterprises in 2015 or Tata Motors in 2025), so no indicator or label crosses an artificial price jump; segments shorter than 300 days were discarded. Rows without a full year of history (needed for the 200-day average and its slope) were removed. The final dataset has 741,980 stock-days for 191 stocks from 19 October 2007 to 13 October 2026, of which 740,519 have a complete label. NIFTY 50 data starts in 2012 and India VIX in 2014; earlier rows keep those features as missing values, which the model handles natively.

# 5. Methodology

## 5.1 Problem formulation and labelling

For a stock on day t, a hypothetical long trade is opened at the open of day t+1. Its take-profit is 0.75 x ATR% and its stop-loss is 4 x ATR% below entry, where ATR% is the 14-day Average True Range as a percentage of price on day t (clamped to 0.8% to 6% and 2% to 15%). The trade is simulated day by day on daily highs and lows: if a day touches both levels the stop is assumed to be hit first (pessimistic), gaps through a level fill at the open, and if neither level is hit by day 15 the trade exits at that day's close. The net return subtracts the full Fyers round-trip cost for a Rs. 50,000 delivery trade (0.446% including 0.05% slippage per side). The binary label is 1 if the net return is positive. This is a form of the triple-barrier labelling method described by Lopez de Prado.

Asymmetric barriers raise the natural win rate (77.1% if one simply bought every day) but losses are larger than wins, so buying every day loses money (average -0.13% per trade). The model's job is to find the minority of setups where the win rate is high enough to make the expected value clearly positive.

## 5.2 Features

All 57 features are percentages, ratios or bounded oscillators so that stocks at different price levels are comparable: 39 stock features (trend distances to EMA 9/20/21 and SMA 20/50/200, SMA 50 vs 200 gap and SMA 200 slope, RSI 2/5/14, MACD and histogram as % of price, Stochastic, Williams %R, CCI, ADX and DI difference, returns over 1/3/5/10/20/60 days, Bollinger %B and width, ATR%, volatility and its ratio, volume ratio, rolling VWAP gap, OBV change, internal bar strength, opening gap, down-streak, distance from 5-day low/high, 20-day z-score, distance from 52-week high/low), 9 market features (NIFTY 50 momentum, distance from its 200-day average, NIFTY RSI, India VIX level and change, stock-vs-index relative strength) and 9 cross-sectional features (each stock's percentile rank among all stocks that day on seven measures, plus two market-breadth ratios).

### 5.3 Model selection

Version 1 compared logistic regression, random forest, extra trees and gradient boosting on 3,776 rows; histogram gradient boosting was best. Version 2 keeps scikit-learn's HistGradientBoostingClassifier because it trains quickly on 740,000 rows, captures non-linear interactions between indicators, and handles missing values without imputation. Three such models with different random seeds and 60% feature subsampling are averaged (bagging), which reduced noise in the high-confidence tail and raised the out-of-sample profit factor in our experiments. Hyperparameters: 300 iterations, learning rate 0.05, 31 leaves, at least 400 samples per leaf, L2 regularisation 1.0.

### 5.4 Walk-forward validation and threshold selection

For each test year Y from 2018 to 2026, a model is trained only on rows dated at least 30 days before 1 January of Y (an embargo longer than the 15-day label horizon, so no label overlaps the test year) and then scores every stock-day of Y. Because the overall level of model scores drifts with market regime, a fixed probability cut-off did not transfer between years (a cut-off chosen on 2018 to 2021 produced only 3 trades in 2022 to 2026). The bot therefore buys when a score is in the top 0.3% of all scores from the previous 250 trading days, a rolling and fully causal threshold. The quantile (0.997) was chosen using only the 2018 to 2021 out-of-sample trades, as the most active setting that reached a 92% win rate and a profit factor of at least 1.5 there. 2022 to October 2026 was then used as the untouched holdout.

### 5.5 Trade gate and portfolio rules

Analysis of losing trades showed they clustered in falling stocks and panicky markets. A rule-based gate therefore only allows trades when the stock's 200-day average is rising, price is no more than 10% below it, ATR is below 4% and India VIX is below 30. The model is trained on all rows (training only inside the gate performed worse) and the gate filters what it may buy. The portfolio rules mirror the live bot: one position per stock, at most 3 new trades per day (highest scores first) and at most 10 open positions.

## 6. Implementation Details

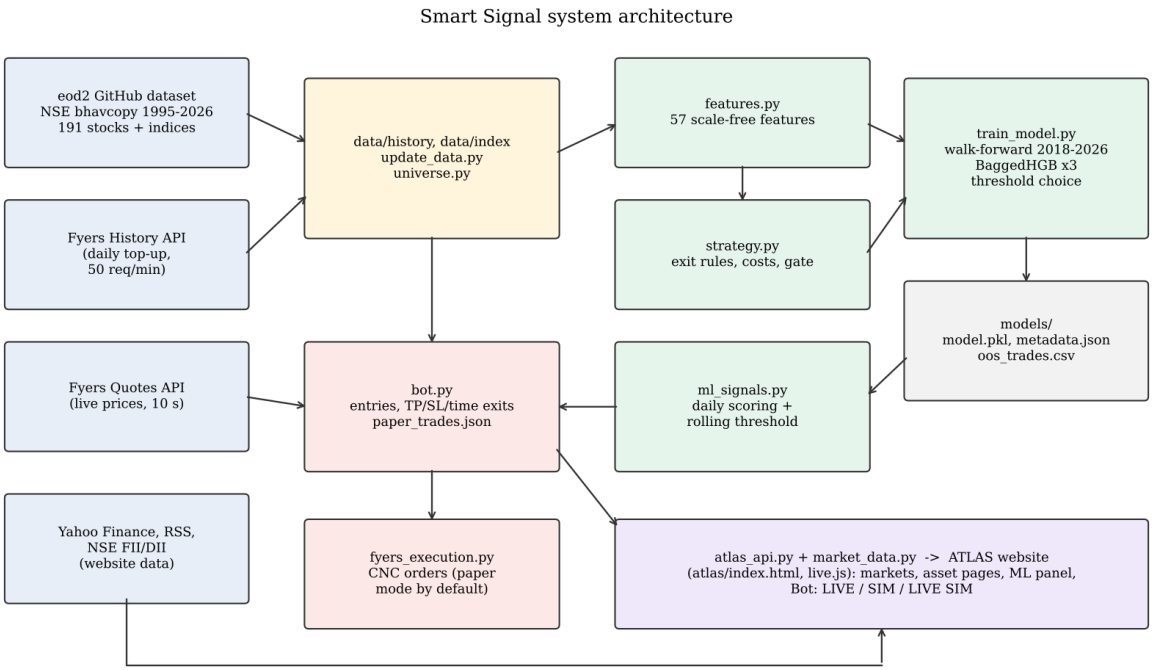



*Figure 1. System architecture.*

| Component | Implementation |
|---|---|
| Language and libraries | Python 3.12, pandas, NumPy, scikit-learn 1.9, joblib, Flask, fyers-apiv3, yfinance, matplotlib; JavaScript and HTML for the front end |
| Data layer | universe.py (symbols, loading, segmenting), update_data.py (Fyers top-up paced under the 50 requests/minute plan limit, GitHub fallback) |
| Features and labels | features.py (vectorised indicators, market and cross-sectional features), strategy.py (exit rules, cost model, gate, trade simulator) |
| Training | train_model.py (dataset build, walk-forward, threshold choice, final model), ensemble.py (BaggedHGB) |
| Evaluation | backtest.py (yearly results, cost sensitivity, NIFTY benchmark), report charts in reports/ |
| Live inference | ml_signals.py (scores the latest completed daily bar of every stock once per day, applies the gate, keeps the rolling threshold up to date) |
| Trading bot | bot.py (morning update and scoring from 08:30, entries until 11:00, take-profit/stop-loss checks every 10 seconds, time stop at 15:20, positions saved to paper_trades.json, automatic reload of the daily Fyers token) |
| Execution | fyers_execution.py (CNC delivery orders; LIVE_TRADING switch off by default, so all orders are paper trades; kill-switch on daily loss and trade count) |
| Web application | atlas_api.py and market_data.py (Flask API with caching and failure back-off), atlas/index.html and atlas/live.js (ATLAS front end) |
| SIM engine | replay.py (replays the 133 real out-of-sample trades on real daily NSE prices, one trading day per 45 seconds, resumable) |
| LIVE SIM engine | sim_market.py (a simulated 8-stock market generating one 5-minute candle per second, 24/7, and the bot trading it with the strategy's gate and exit rules) |

The ATLAS website, served at http://127.0.0.1:5000, shows global indices, Indian and US stocks, forex, commodities, crypto, India sector and breadth data, news and a stock screener.

Every NSE stock page shows the model's estimated probability of a winning trade, whether the stock passes the gate and, for a qualifying setup, the planned entry, target and stop. The Bot section ('Today's Performance' on the home page and a full Trading bot page) has three modes, described below.

### 6.1 Bot modes: LIVE, SIM and LIVE SIM

**LIVE** shows the running bot (bot.py) on the real market: Fyers quotes every 10 seconds, the day's model picks, open positions with their target and stop, the order log and today's P&L. Because the strategy buys only the top 0.3% of setups, on many days it correctly buys nothing.

**SIM** is the evidence view. replay.py plays the model's real walk-forward trades back in real time: each historical trading day lasts 45 seconds, prices tick along a path through that day's real open, high, low and close, BUY orders fire at the real entry price and SELL orders fire when the price reaches the trade's real exit level. It starts with the 2022 to 2026 holdout, continues with 2018 to 2021 and loops. Every number shown is a real out-of-sample result after charges; below the replay the page summarises all 133 trades and the equity curve.

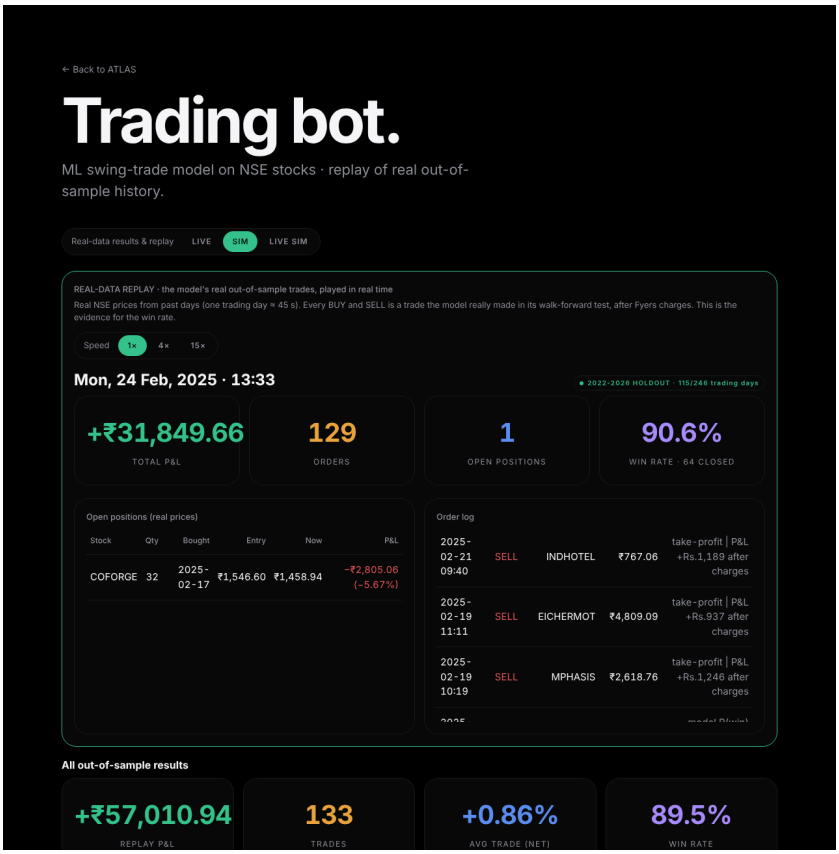



*Figure 2. SIM tab: real out-of-sample trades replayed in real time on real NSE prices.*

**LIVE SIM** lets the bot be demonstrated when the NSE is closed. sim_market.py generates a market for eight stocks (HDFCBANK, RELIANCE, AUBANK, BAJAJFINSV, ICICIBANK, INFY, TCS and SBIN) in real time, one 5-minute candle per second, 24 hours a day. No past prices are replayed; only each stock's starting price is taken from its latest real close. Each simulated stock follows a slowly switching trend regime (up, flat or down), mean reversion

towards a drifting fair value, clustered volatility and rare news jumps. On every candle the bot computes indicators from the chart (SMA 50 and 200 and the 200-bar slope, RSI 14, RSI 2, 20-bar z-score and ATR), applies the same trend and volatility gate as the real strategy, buys when a stock in an uptrend dips sharply (RSI 2 below 5 and price more than 2 standard deviations below its 20-bar mean), and sells at the strategy's take-profit (0.75 x ATR), stop-loss (4 x ATR) or after 30 candles. Each trade is Rs. 50,000 and pays the Fyers cost model. The page shows live charts with buy and sell markers, open positions, the order log, P&L and win rate, with 1x, 4x and 10x speed.

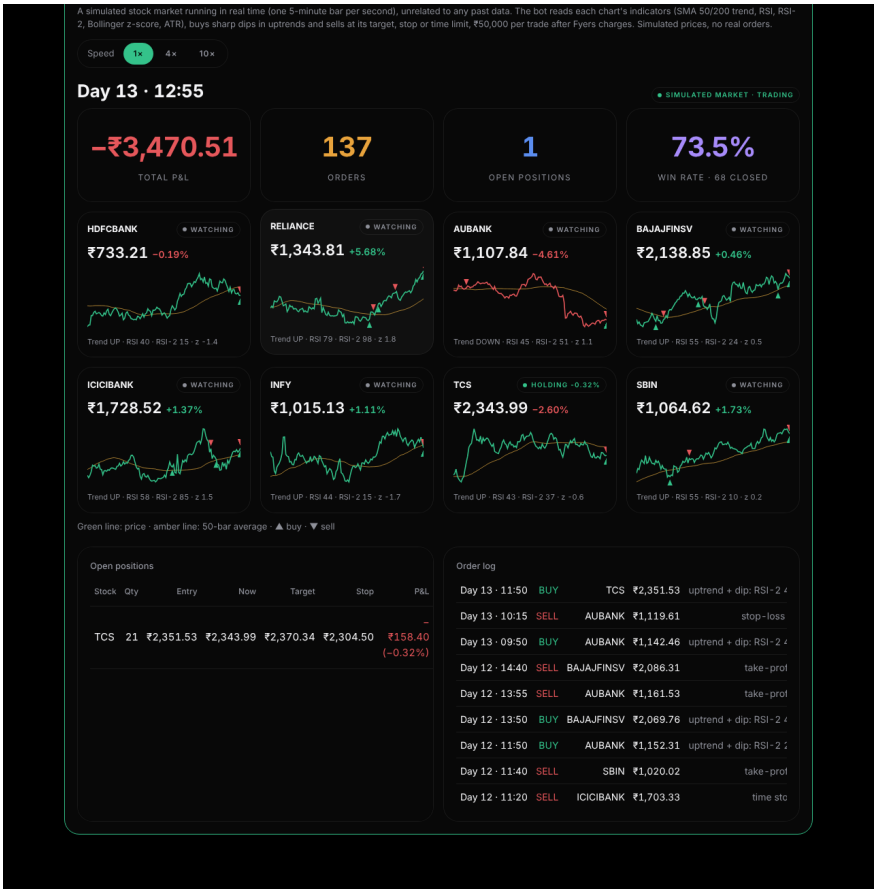



*Figure 3. LIVE SIM tab: the bot trading the simulated market, with charts, positions and order log.*

The LIVE SIM market's parameters were fixed in advance and were not tuned to flatter the bot. Over five independent runs of 20 simulated days each (487 closed trades) the bot won 81.5% of trades with an average net result of about -Rs. 27 per trade, i.e. roughly break-even. LIVE SIM therefore demonstrates that the bot reads charts and places and closes trades on its own in real time; it is not evidence of the model's accuracy, which comes only from the real-data walk-forward results in Section 7 and the SIM tab.

# 7. Results and Discussion

## 7.1 Main results

| Period (all out-of-sample) | Trades | Win rate | Avg net / trade | Profit factor | P&L at Rs. 50,000 / trade |
|---|---|---|---|---|---|
| 2018 to 2021 (used to choose the threshold) | 46 | 93.5% | +1.04% | 2.27 | Rs. 24,012 |
| **2022 to Oct 2026 (holdout)** | **87** | **87.4%** | **+0.76%** | **2.04** | **Rs. 32,999** |
| All 2018 to 2026 | 133 | 89.5% | +0.86% | 2.12 | Rs. 57,011 |
| Version 1 model (for comparison) | n/a | 37.3% accuracy (3-class) | n/a | n/a | n/a |

Exits: 119 take-profit, 9 time stops, 5 stop-losses. Average holding period 4.2 trading days. Winning trades averaged +1.8% and losing trades -7.2% net. The 95% confidence interval for the 89.5% win rate on 133 trades is roughly 84% to 95%.

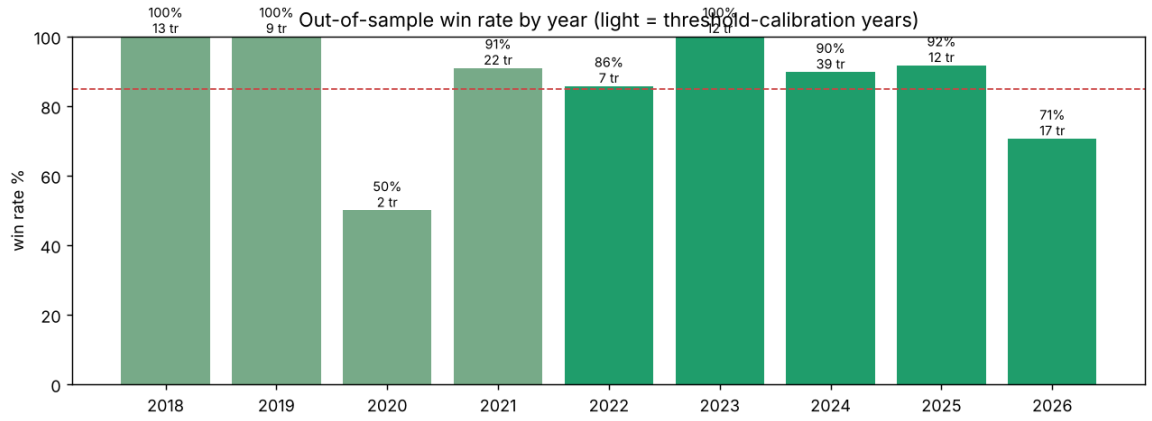



*Figure 4. Out-of-sample win rate per year (red line at 85%).*

| Year | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 |
|---|---|---|---|---|---|---|---|---|---|
| Trades | 13 | 9 | 2 | 22 | 7 | 12 | 39 | 12 | 17 |
| Win rate | 100% | 100% | 50% | 91% | 86% | 100% | 90% | 92% | 71% |

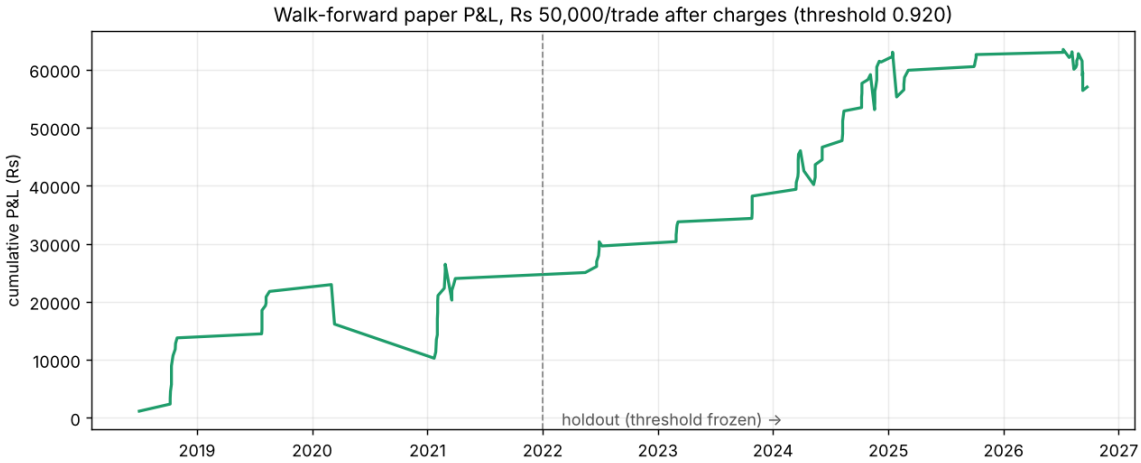



*Figure 5. Cumulative walk-forward P&L at Rs. 50,000 per trade after charges.*

## 7.2 Analysis of model quality

Across all 392,537 out-of-sample stock-days the model's ROC AUC is only about 0.525 (between 0.48 and 0.57 per year), so its ability to rank ordinary days is weak, which is expected for daily stock returns. The useful signal lies in the extreme tail: among stock-days in the top 0.3% of scores that also pass the gate, 85.2% won, against a 77% base rate, and the portfolio rules (rolling threshold, best 3 per day, one position per stock) raise this to 89.5% on the trades actually taken. Figure 6 shows that higher confidence bands outside the gate do not achieve this, which is why both the model and the gate are needed.

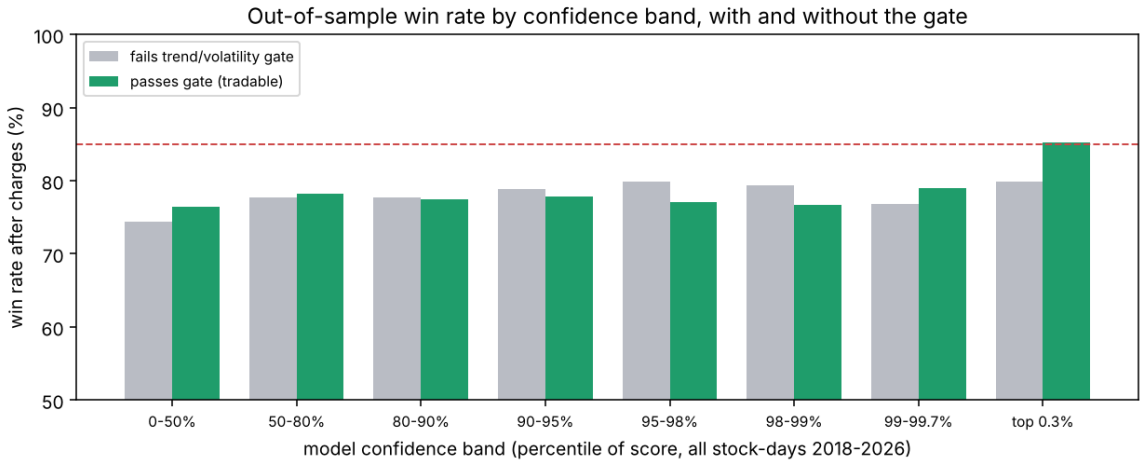



*Figure 6. Out-of-sample win rate by model confidence band, inside and outside the gate.*

### 7.3 Effect of trading costs

| Rs. per trade | Round-trip charges | Win rate | Avg net / trade | Profit factor |
|---|---|---|---|---|
| 5,000 | 1.33% | 85.0% | -0.02% | 0.97 |
| 10,000 | 0.94% | 88.7% | +0.36% | 1.44 |
| 25,000 | 0.57% | 89.5% | +0.73% | 1.95 |
| 50,000 | 0.45% | 89.5% | +0.86% | 2.12 |
| 1,00,000 | 0.38% | 89.5% | +0.92% | 2.22 |

Fyers charges a flat Rs. 20 per order and Rs. 14.75 per sell for depository fees, so small trades lose most of the edge. At Rs. 5,000 per trade the same signals are slightly loss-making.

### 7.4 Novelties

- Labels that match the executed strategy, including the exact Fyers fee schedule, instead of generic up/down labels.
- A rolling, causal score-quantile threshold that adapts to regime drift, chosen only on calibration years.
- Cross-sectional rank and breadth features computed across the whole universe each day, both in training and live.
- A complete working system: live Fyers paper trading with token hot-reload, rate-limit pacing and persistent positions.
- The ATLAS dashboard with a per-stock model panel, a SIM mode that replays the model's real out-of-sample trades in real time, and a 24/7 LIVE SIM mode in which the bot trades a simulated market for demonstrations.

### 7.5 Limitations

- Not every year reaches 85%: 2020 (2 trades, 50%) and 2026 so far (17 trades, 70.6%) fall short.
- Weak overall discrimination (AUC about 0.53); the edge depends on a small number of very selective trades.
- Infrequent trading (about 16 trades a year) and modest total profit: Rs. 57,011 over 8.2 years would have needed up to Rs. 5 lakh reserved (about 1.3% per year), while NIFTY 50 buy-and-hold returned about 10% per year.
- Survivorship bias, because the universe is today's large companies.
- The design (exit rules, gate, bagging, threshold target) was explored on the same history, so the holdout is strong but not perfectly untouched evidence.
- Fills at the open and at exact target/stop levels are assumed; slippage beyond 0.05% per side is not modelled.
- LIVE SIM uses a simplified rule (dip in an uptrend) on synthetic intraday candles rather than the daily ML model, and in that synthetic market the win rate is about 81% and P&L about break-even, so it is a demonstration only.

# 8. Conclusion and Future Scope

Reformulating the problem from three-class direction prediction to cost-aware trade-outcome classification, enlarging the dataset about 200-fold and validating strictly walk-forward turned a model with 37% accuracy and no working exits into a system that won 87.4% of its trades after charges on the 2022 to 2026 holdout, with a profit factor above 2. The project also delivers a working paper-trading bot and an integrated dashboard. The main lesson is that a high win rate is achievable only through selectivity, and must always be read together with profit factor, trade frequency and costs.

**Future scope:** intraday entry timing to reduce gap risk; survivorship-free index membership history; probability calibration and conformal methods for the threshold; position sizing by confidence and volatility; short-selling and sector-neutral variants; regime detection to pause trading in years like 2020; and several months of forward paper trading before any decision to use real money.

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